

Supplementary Material

Full derivations and diagnostic case study for

“What can media-only liver-on-a-chip concentration data identify?”

Structural identifiability, a continuous equivalence family, and a cautionary case study in practical identifiability”

This document gives the full derivations behind the Theoretical section of the main paper, and documents in detail the sequence of diagnostic attempts summarized qualitatively in the Discussion.

S1. Model equations

Two-compartment model: media compartment (concentration C_m , volume V_m) and a lumped cell-associated compartment (concentration C_c , volume V_c), linked by distributional clearance CL_d and cell:media partition coefficient K_p , first-order non-specific binding (NSB) loss from media at rate k_b , and metabolism (intrinsic clearance CL_{int}) confined to the cell compartment:

$$\frac{dC_m}{dt} = -\frac{CL_d}{V_m}C_m + \frac{CL_d}{K_p V_m}C_c - k_b C_m, \quad (1)$$

$$\frac{dC_c}{dt} = \frac{CL_d}{V_c}C_m - \left(\frac{CL_d}{K_p V_c} + \frac{CL_{int}}{V_c} \right) C_c. \quad (2)$$

Matrix form $\dot{\mathbf{x}} = A\mathbf{x}$ with $\mathbf{x} = [C_m, C_c]^T$ and

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad a_{11} = -\left(\frac{CL_d}{V_m} + k_b \right), \quad a_{12} = \frac{CL_d}{K_p V_m}, \quad a_{21} = \frac{CL_d}{V_c}, \quad a_{22} = -\left(\frac{CL_d}{K_p V_c} + \frac{CL_{int}}{V_c} \right).$$

Media volume with evaporation: $V_m(t) = \max(V_{m0} - e \cdot t, \text{floor})$, with e a linear-in-time evaporation rate. Evaporation is treated as a known, externally measured input (via residual-volume weighing) throughout the structural identifiability derivation in S2; the full simulation code allows e to be either fixed or jointly fitted, and the distinction matters for the practical-identifiability case study in S4. Initial condition: dose in media only, $C_m(0) = C_0$, $C_c(0) = 0$.

S2. Laplace-transform derivation of structural identifiability

Taking the Laplace transform of the system above with $y(t) = C_m(t)$ observed and $\mathbf{x}(0) = [C_0, 0]^T$:

$$\frac{Y(s)}{C_0} = \frac{s - a_{22}}{s^2 - (a_{11} + a_{22})s + (a_{11}a_{22} - a_{12}a_{21})}.$$

The denominator and numerator coefficients are the only quantities determined by data of arbitrary density and duration:

- Numerator constant term: $-a_{22}$

- Denominator s^1 coefficient: $-(a_{11} + a_{22})$, hence a_{11} given a_{22}
- Denominator s^0 coefficient: $a_{11}a_{22} - a_{12}a_{21}$, hence $a_{12}a_{21} \equiv P = CL_d^2/(K_p V_m V_c)$ given a_{11} , a_{22}

So exactly three real numbers — a_{11} , a_{22} , P — are identifiable from $C_m(t)$, regardless of sampling. With V_m , V_c known, this gives 3 equations in the 4 unknowns CL_d , K_p , CL_{int} , k_b :

$$a_{11} = -\left(\frac{CL_d}{V_m} + k_b\right), \quad a_{22} = -\left(\frac{CL_d}{K_p V_c} + \frac{CL_{int}}{V_c}\right), \quad P = \frac{CL_d^2}{K_p V_m V_c}.$$

S2.1 Continuous equivalence family

Solving for (CL_d, K_p, CL_{int}) in terms of a free parameter k_b :

$$CL_d(k_b) = V_m(-a_{11} - k_b), \quad (3)$$

$$K_p(k_b) = \frac{CL_d(k_b)^2}{P V_m V_c}, \quad (4)$$

$$CL_{int}(k_b) = -a_{22}V_c - \frac{CL_d(k_b)}{K_p(k_b)}. \quad (5)$$

Any k_b in the admissible range (yielding positive CL_d , K_p , CL_{int}) generates a full parameter quadruple consistent with the same (a_{11}, a_{22}, P) , hence the same $C_m(t)$ trajectory exactly. This was verified symbolically (SymPy): substituting the closed-form solution back into the definitions of a_{11} , a_{22} , P recovers the assumed values identically for arbitrary k_b .

Numerical verification. Two parameter sets — $(CL_{int}, CL_d, K_p, k_b) = (4.5, 8.0, 5.0, 0.0008)$ and $(3.243, 4.48, 1.568, 0.003)$, both consistent with the same (a_{11}, a_{22}, P) by construction — were each simulated through the full nonlinear ODE solver (constant-volume case, $e = 0$) and their $C_m(t)$ trajectories compared. Maximum absolute difference: 3.76×10^{-8} ; maximum relative difference: 3.92×10^{-7} (attributable to `solve_ivp` numerical tolerance, not a real difference — the trajectories are analytically identical).

S2.2 Resolution via independent k_b

If k_b is measured independently, the three equations above become 3 equations in the 3 remaining unknowns (CL_d, K_p, CL_{int}) , with a unique solution given by the same closed-form expressions evaluated at the known k_b .

S3. Exact asymptotic bias limits

Define the conventional apparent clearance as $CL_{naive} = V_m \cdot k_{dep}$, where k_{dep} is the monoexponential media-depletion rate constant obtained by log-linear regression of $C_m(t)$ assuming a single well-mixed compartment at constant V_m . The terminal decay rate an analyst would fit corresponds to the eigenvalue of A with smaller magnitude (λ_{slow}), so $CL_{naive} = -\lambda_{slow} \cdot V_m$.

S3.1 Fast-distribution limit ($CL_d/CL_{int} \rightarrow \infty$)

As $CL_d \rightarrow \infty$ with the other parameters fixed, λ_{slow} converges to

$$\lambda_{slow} \rightarrow -\frac{K_p CL_{int} + k_b V_m}{V_m + K_p V_c}, \quad \text{so} \quad CL_{naive} \rightarrow \frac{V_m(K_p CL_{int} + k_b V_m)}{V_m + K_p V_c}.$$

Confirmed numerically via direct eigenvalue evaluation across $CL_d \in \{8, 80, 800, \dots, 8 \times 10^{12}\} \mu\text{L}/\text{min}$ (other parameters: $CL_{int} = 4.5$, $K_p = 5.0$, $k_b = 0.0008$, $V_m = 1600$, $V_c = 3.0$; asymptotic value 23.5591):

CL_d	CL_{naive}
8	7.177
80	18.734
800	22.961
8,000	23.498
8×10^5	23.5585
8×10^8	23.5591

S3.2 Fast-metabolism limit ($CL_{int}/CL_d \rightarrow \infty$)

As $CL_{int} \rightarrow \infty$, the cell compartment approaches a quasi-steady-state sink ($C_c \rightarrow 0$ relative to C_m), and $\lambda_{slow} \rightarrow -(CL_d/V_m + k_b)$, giving $CL_{naive} \rightarrow CL_d + k_b V_m$, independent of CL_{int} . Confirmed numerically across $CL_{int} \in \{4.5, 45, \dots, 4.5 \times 10^{10}\}$ ($CL_d = 8.0$ fixed): CL_{naive} converges to 9.28 ($= 8.0 + 1600 \times 0.0008$) by $CL_{int} = 4500$, matching to 4 significant figures.

S3.3 Nondimensionalization

With $\tau = (CL_d/V_m)t$, $x = C_m/C_m(0)$, $y = C_c/(K_p C_m(0))$:

$$\frac{dx}{d\tau} = -(1 + \beta)x + y, \quad \frac{dy}{d\tau} = \frac{1}{r}[x - (1 + \mu)y],$$

where $r = K_p V_c/V_m$, $\beta = k_b V_m/CL_d$, $\mu = K_p CL_{int}/CL_d$. Derived and verified symbolically (chain-rule substitution of $C_m = C_m(0) \cdot x(\tau)$, $C_c = K_p C_m(0) \cdot y(\tau)$, $t = (V_m/CL_d)\tau$ into the original ODEs). Rewriting the two asymptotic bias-ratio limits in these dimensionless terms:

$$\frac{CL_{naive}}{CL_{int}} \rightarrow \frac{K_p}{1+r} \left(1 + \frac{\beta}{\mu}\right) \quad \text{as } CL_d/CL_{int} \rightarrow \infty, \quad \frac{CL_{naive}}{CL_{int}} \rightarrow \frac{K_p}{\mu} (1 + \beta) \quad \text{as } CL_{int}/CL_d \rightarrow \infty.$$

Note that $\mu \rightarrow 0$ or $\mu \rightarrow \infty$ alone does not uniquely specify which physical limit is meant ($\mu = K_p CL_{int}/CL_d$ can approach these limits via $CL_d \rightarrow \infty$, $CL_{int} \rightarrow 0$, or $K_p \rightarrow 0/\infty$ separately); the two formulas above are specifically derived along the $CL_d \rightarrow \infty$ and $CL_{int} \rightarrow \infty$ paths respectively, with the other parameters held fixed.

S4. Practical identifiability: a methodological case study

This section documents in full the sequence of diagnostic attempts summarized qualitatively in the Discussion.

S4.1 FIM covariance approximation: numerical instability near the structural degeneracy

For representative scenarios (CN Bio hardware, $V_m = 1600 \mu\text{L}$, $V_c = 3 \mu\text{L}$, low- and high- CL_{int} regimes, dense [$n = 12$] sampling, 15% proportional noise), nonlinear least-squares fits (trust-region-reflective) were performed with all 5 parameters (CL_{int} , CL_d , K_p , k_b , e) free. The Gauss-Newton Fisher Information Matrix $J^T J$ was computed at each fitted optimum.

Observed condition numbers of $J^T J$: 1.377×10^{17} (low- CL_{int}) and 1.447×10^{12} (high- CL_{int}) — both indicating numerical singularity at or beyond double-precision limits. Comparing plain matrix inversion against a truncated-SVD pseudoinverse ($\text{rcond} = 10^{-10}$) on the identical fitted Jacobian: relative standard errors for CL_{int} of 1.404×10^3 (plain inverse) vs. 6.885 (pseudoinverse) for the low- CL_{int} scenario, and 2.367 vs. 0.190 for the high- CL_{int} scenario — disagreements of two to three orders of magnitude from the choice of inversion algorithm alone, on identical data and an identical point estimate.

S4.2 Single-start profile likelihood: contaminated by convergence failure

A profile likelihood for CL_{int} (fixing CL_{int} at a grid of values relative to its fitted value, re-optimizing all other free parameters, recording the resulting SSE) was computed with a single optimizer start per grid point. The resulting profile was non-monotonic and erratic: near-zero ΔSSE (as good as the unconstrained optimum) recurred at several grid points interspersed with a sharp spike ($\Delta\text{SSE} = 0.34$) at an intermediate ratio ($4.8\times$) — a pattern inconsistent with a well-behaved likelihood profile and diagnostic of the optimizer failing to locate the true conditional optimum at that grid point from a fixed, non-adaptive starting guess.

S4.3 Multi-start profile likelihood

Repeating the profile at each grid point with 6 random restarts (starting multipliers drawn uniformly from $[0.3, 3.0] \times$ the nominal parameter values) and retaining the best (lowest-SSE) fit at each point gave materially more stable profiles. Three findings:

1. k_b , e **unknown, sparse sampling** ($n = 5$): ΔSSE remained near zero (all 7 scanned points inside a 95% profile-likelihood confidence region, χ_1^2 threshold) across the full scanned range of CL_{int} ratios ($0.3\times$ – $3.0\times$), i.e. CL_{int} is not practically constrained by this design.
2. k_b , e **known (fixed at true values), sparse sampling** ($n = 5$): ΔSSE also remained near zero across the same range — fixing k_b and e did not, in this regime, materially sharpen the practical identifiability of CL_{int} .
3. k_b , e **known, dense sampling** ($n = 12$): the same result held even under dense sampling.

This is the basis for the main text’s methodological finding: resolving the structural degeneracy (S2.2) does not by itself guarantee practical estimability under realistic finite, noisy sampling designs, in the parameter regime examined (low CL_{int} relative to CL_d , i.e. small μ).

S4.4 Scope and caveats

All numerical values above are for one illustrative parameter regime (CN Bio hardware, low- CL_{int} level) and should not be read as general claims about all regimes or platforms. The multi-start profile likelihood used 6 restarts per grid point with uniform-multiplier initialization; a more exhaustive search might further change the profile shape at the extremes of the scanned range. We did not perform a DAISY-based or other formal structural identifiability check on the complete 5-parameter, time-varying-volume model; it is possible additional structural degeneracies exist between k_b and e in the full model.

S5. Code and reproducibility

All derivations, symbolic verification, and numerical checks were performed in Python 3 (NumPy, SciPy, SymPy, Matplotlib). Code files:

- `model.py` — ODE right-hand side, platform hardware parameters, media-concentration simulation
- `analysis.py` — synthetic data generation, naive and mechanistic fitting, FIM-based practical identifiability
- `structural_identifiability.py` — symbolic and numerical verification of the equivalence family (S2)
- `profile_likelihood.py`, `diagnostics_S4.py` — profile-likelihood scanning utilities and the consolidated, end-to-end script reproducing every numbered claim in S4.1–S4.3
- `make_results_figures.py`, `build_figure1.py` — figure-generation scripts for all main-text figures

References

- [1] Docci L, Milani N, Ramp T, et al. Exploration and application of a liver-on-a-chip device in combination with modelling and simulation for quantitative drug metabolism studies. *Lab Chip*. 2022;22(6):1187–1205.
- [2] Rajan SAP, Sherfey J, Ohri S, et al. A novel milli-fluidic liver tissue chip with continuous recirculation for predictive pharmacokinetics applications. *AAPS J*. 2023;25(6):102.